Proposed "generalization" - LQ Bewley.

guess:

$$b(X_t, y(z_t)) = G(I-\beta A)^{-1} X_t - G_c (I-\beta A)^{-1} X_t$$

depends only on z_0

start with stationary z_0

people initially endowed with $b(z_0, \bar{c})$ claims

$$\int b(z_0, \bar{c})\, dF(z_0) = 0$$

RAT. outcome

stationary dist.

Pure exchange:

$$y_t^i = G_1 x_t + G_2 z_t, \quad \begin{pmatrix} x_{t+1} \\ z_{t+1} \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix} \begin{pmatrix} x_t \\ z_t \end{pmatrix} + \begin{pmatrix} C_{11} & C_{12} \\ 0 & C_{22} \end{pmatrix} \begin{pmatrix} w_{1t+1} \\ w_{2t+1} \end{pmatrix}$$

$$= G S_t$$

$z_t \sim$ aggregate, $x_t \sim$ individual

aggregate: $\bar{y}_t = G_1 m_{x_t} + G_2 z_t$

$m_{x_t} = E\, x_t | z_t = \gamma z_t$, projection

$$\Rightarrow \bar{y}_t = G_1 \gamma z_t + G_2 z_t = G_c S_t$$

$$G_c = (\underbrace{0}_{x_t}, \underbrace{G_1 \gamma + G_2}_{z_t})$$

$$S_t = \begin{pmatrix} x_t \\ z_t \end{pmatrix} \quad \begin{matrix} \text{individual} \\ \text{aggregate} \end{matrix}$$

$$q(S_{t+1} | S_t) = \beta\, \phi(S_{t+1} | S_t)$$

$$\phi \sim \mathcal{N}(A S_t, CC')$$

guess

average consn

$$b(x_t, \bar{y}(z_t)) = \underbrace{G(I-\beta A)^{-1} S_t}_{\text{pv of non-fin income}} - G_c S_t$$

$$c_t(z_t) = \bar{y}_t = G_c S_t$$